

UKAS QUALITY MANAGEMENT 005

SYSTEM CERTIFICATION ISO 9001:2008 SGS

ACROW

Formwork Technology

Aludeck System

ACROW MISR Provides Full Site Service Based On More Than 35 Years Experience In The Field



ACROW

Formwork Technology



Contents

Product Features	2
MIP-Method (Main beam-Infill beam-Plywood)	3
Typical Application	4
PMD-Method (Panel – Main beam – Drop head)	5
PMD Method	6
PMD Method assembly	7
PMD Method Filler Solutions With 18 mm Plywood	9
Panel Method	10
Design tables	11
Components	13

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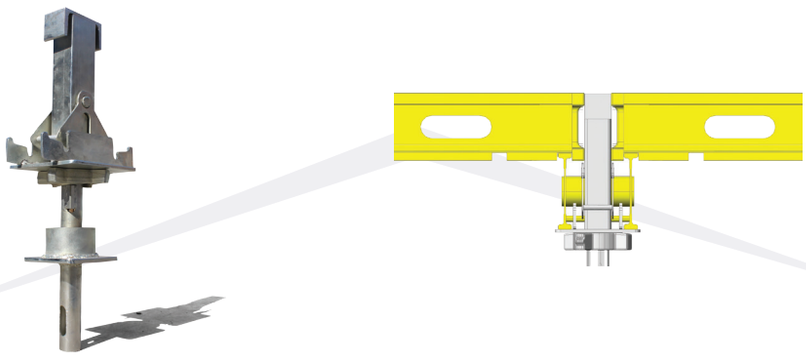
Product Features

The requirements for an economic and safe solution for a slab formwork vary alot in the different construction project.

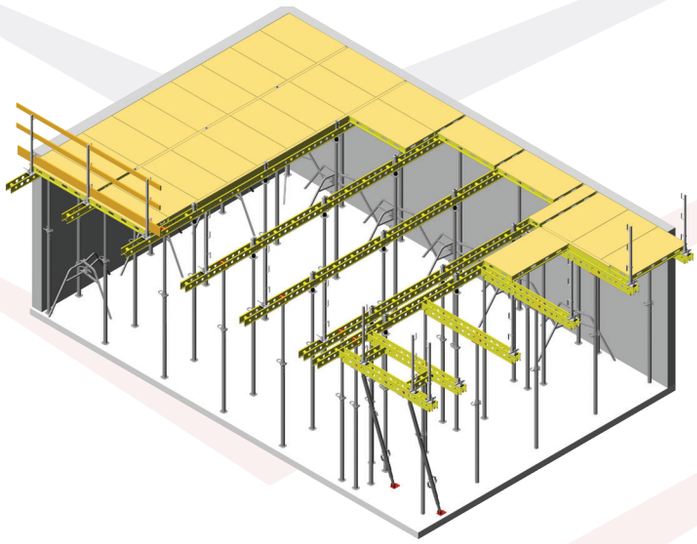
It depends on:
- Floor plan
- Size of the different rooms
- Floor height
- Slab thickness
- Number of reuses
- Concrete surface expected

The conception of the ALUDEC system covers the most important slab form methods just in one system!

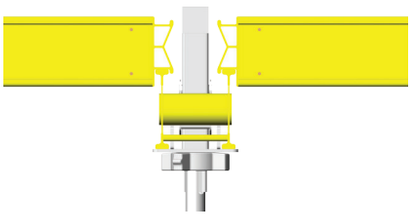
•This means using the same parts in the different applications, also mixing the different Aludec methods-MIP and PMDdepending on the requirements of the floor plan.
•The availability of choosing between separate plywood (MIP method) or panels with implemented plastic–long life-plywood (PMD method).
•The reduction of the labor cost due to the few intensive adjustments on the jobsite.
•The simplicity of the system reduces the mistakes in handling and makes the workers safer.

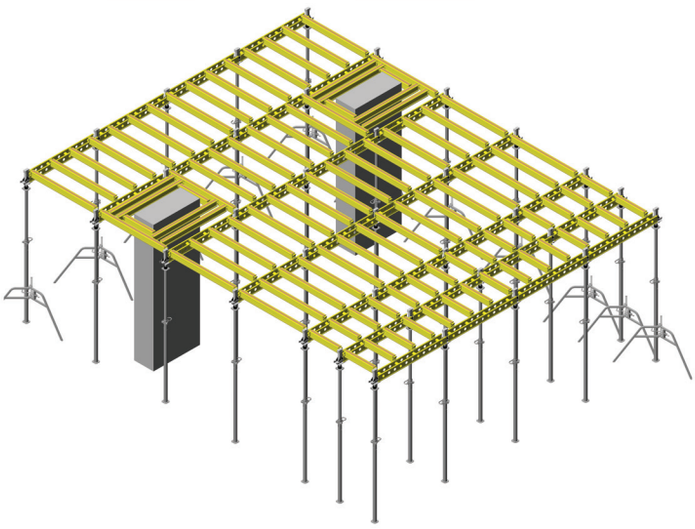


Drophead



PMD method





MIP method

MIP-Method (Main beam-Infill beam-Plywood)

•This method depends on using a mainbeam fit in a drop-head, together with an infill-beam connected in the same level.

•The plywood then can be placed over all parts.

•Infill-beams could be used separately with the twin-head and being fit cross to another infill-beam.

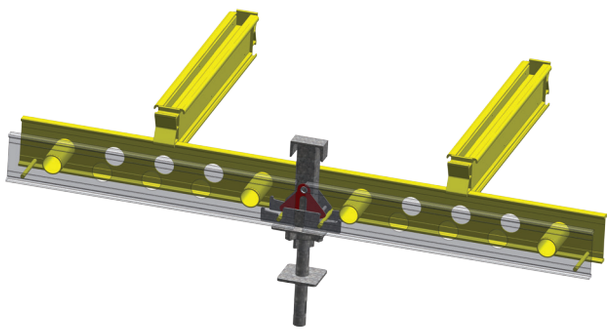
•Drop-Head principle allows striking after 1 day depending on slap thickness and concrete strength.

The main beams have 9 different lengths:

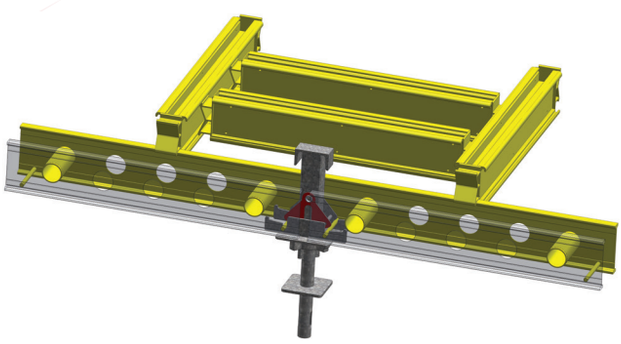
•Main-beam 60 cm, 90 cm, 120 cm, 150cm, 180 cm, 210 cm, 225 cm, 240 cm and 360 cm.

Infill beams have 4 different lengths:

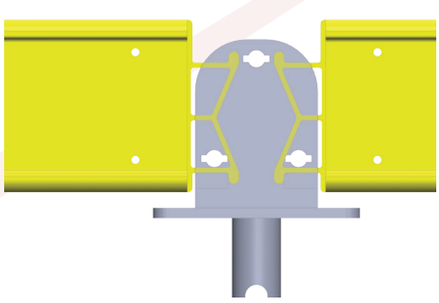
•Infill-beam 90 cm, 120 cm, 150 cm and 180 cm.

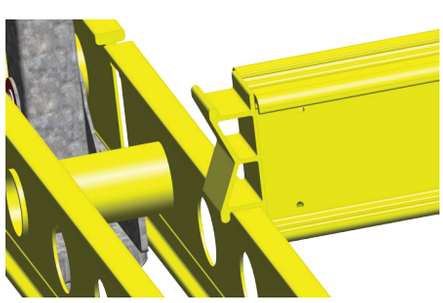


Infill beam fit on the main beam



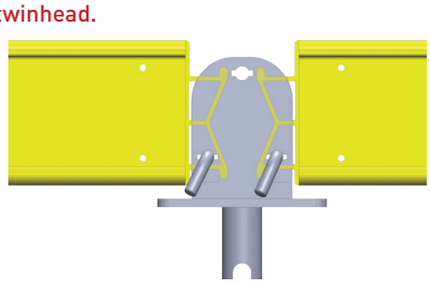
Infill beam fit on on another infill beam for infill solution

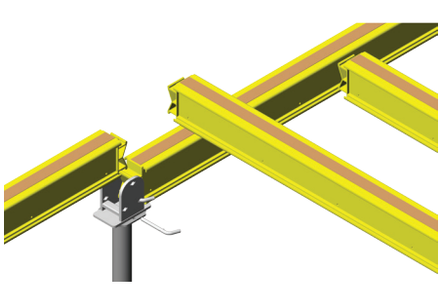


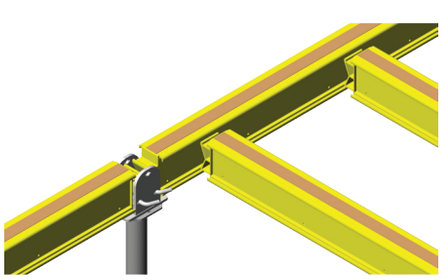


Infill-beam can fit anywhere on the Mainbeamaccording to permissible plywood spacing and according to the load on the infill beam.
Plywood 18 mm is placed on the Infillbeam

Infill-beams are mounted on the twinhead.







Secure the Infill-beams with the safety bolt.

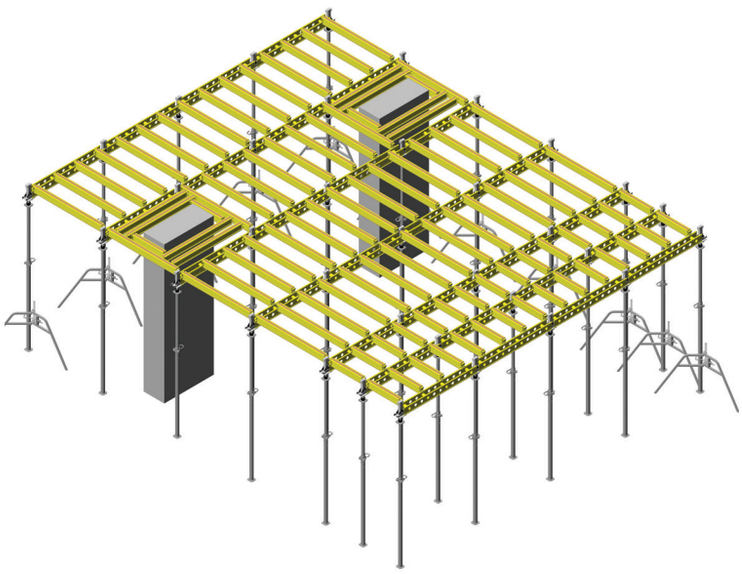
The smart design of the Infill beams allows the main and secondary infill beams to be on the same level.

Typical Application

•Typical application for MIP method where using infill beams with main beams and drop heads.

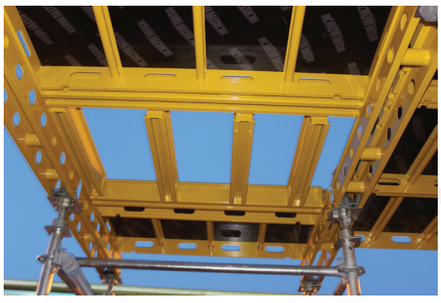
•Different sizes of mean beams could fit through the drop-head.

•The smart design of the drop-head allows 4 cm drop in the system for early striking, which allows reusing the decking beams for the next concreting cycle.



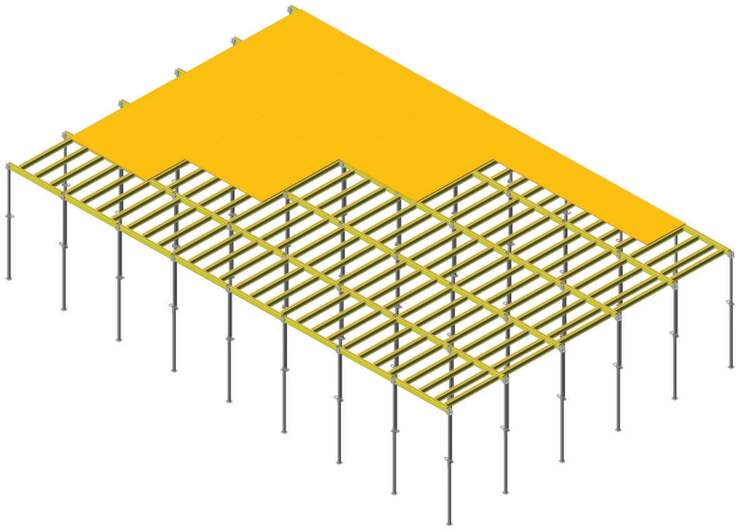
Main beams with Infill beams showing the flexibility of the system and solving the infill areas with the same components

Typical application for MIP method where using Infill beams with infill beams as main and secondary beams.





Infill beam



Only infill beams and twin heads are used, minimizing the components used.

PMD-Method (Panel – Main beam – Drop head)

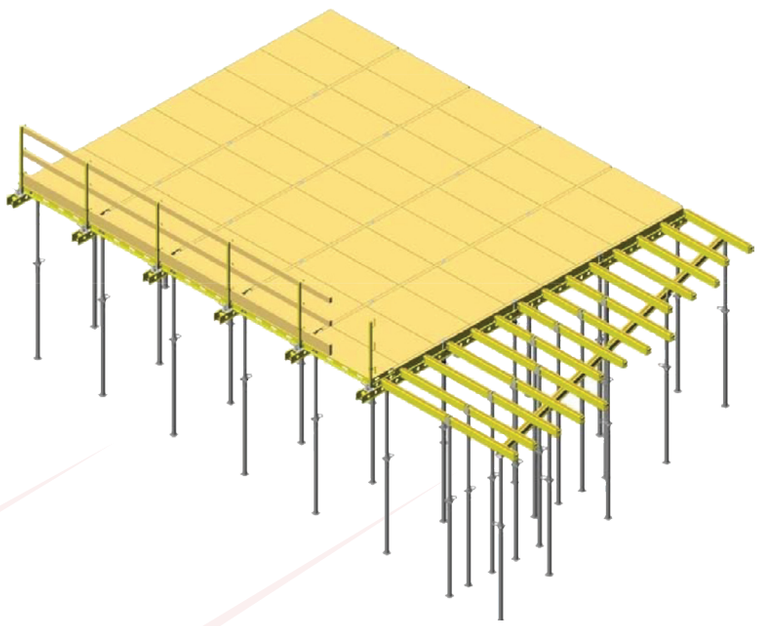
•This method depends on using mainbeams fit in drop-heads-for early strikingthen the panels are placed over the mainbeam.

•The panels are implemented with plywood.

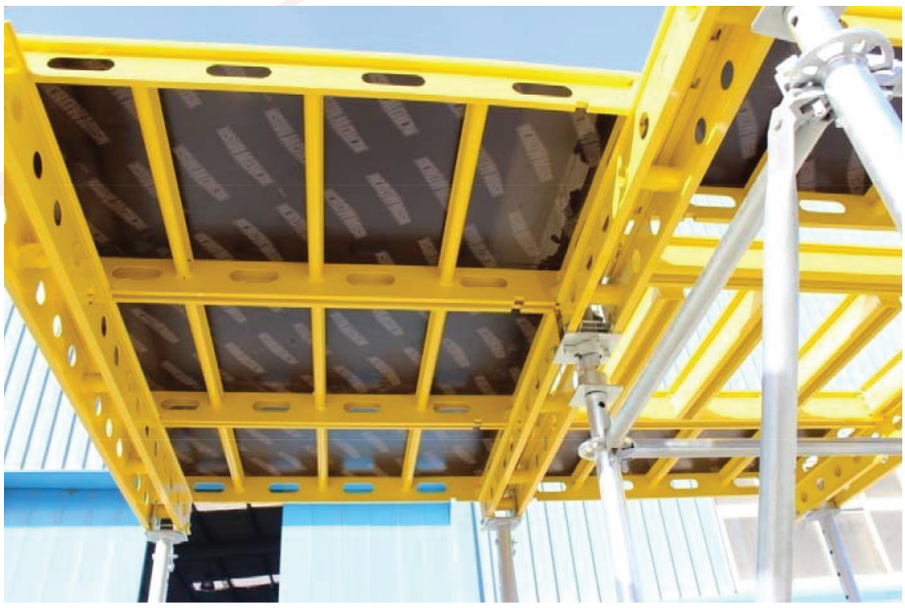
•The panels are placed totally independent from the position of the drop-head and the length of the mainbeam.

•This makes an easy adjustment in the middle of the room without any additional support.

•The number of props is settled by the system, which means no error in handling.







•Early Striking and fast cycling of slab form comes to a reduction of material investment up to 40%, simplify the logistics and man power and reducing the cost accordingly.

•This system can be one of the fastest and cost effective solutions for many projects.

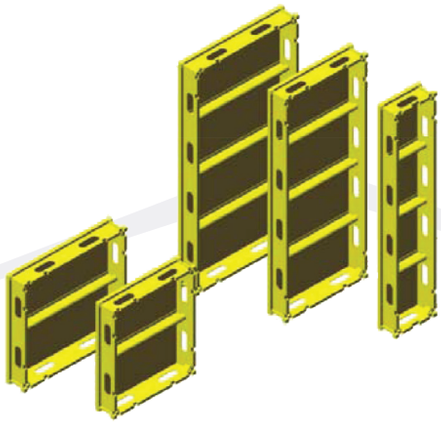
The Aludeck could be used with any of Acrow shoring systems

PMD Method

Panels overview:
The panels have 9 different sizes

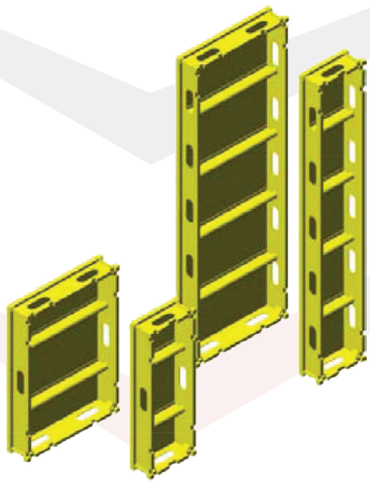
Panels used for slab thickness up to 35 cm :

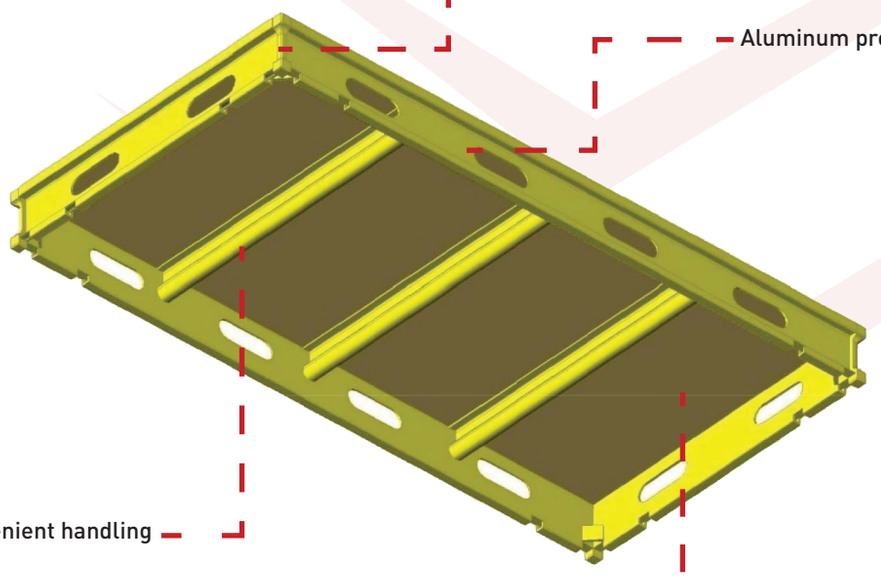
ADP 150 x 75
ADP 150 x 60
ADP 150 x 30
ADP 75 x 75
ADP 75 x 60



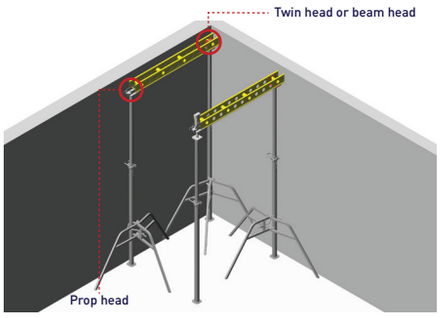
Panels used for slab thickness up to 25 cm :

ADP 180 x 60
ADP 180 x 30
ADP 90 x 60
ADP 90 x 30

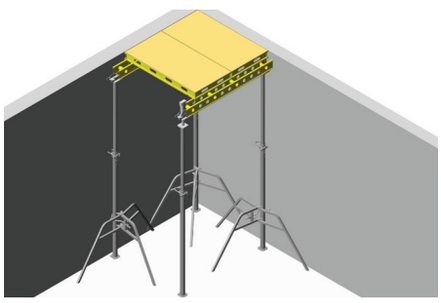




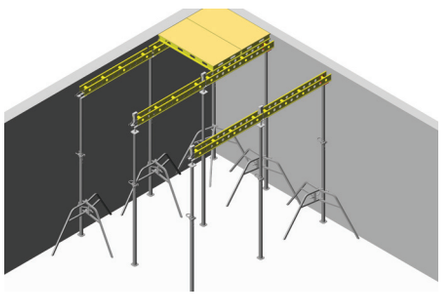
PMD Method assembly



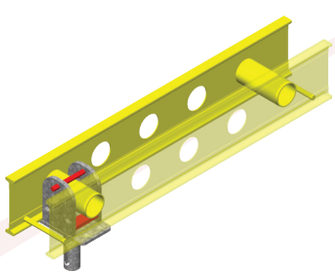
Start in the corner using beam head or twin head with the main beam as in Figure 1.



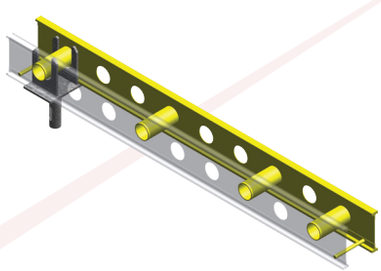
Panels are then fixed easily on the 1st 2 main beams.



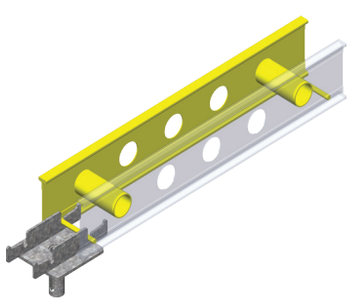
Continue erecting the main beams with the drop heads for early striking.



Twin head secured

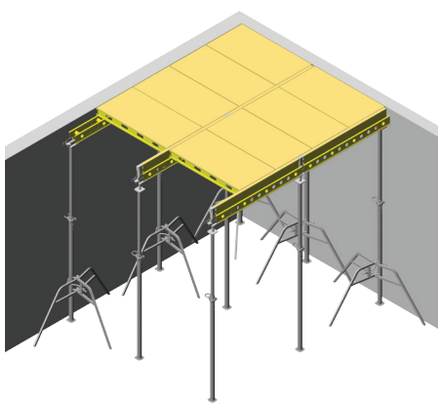


Beam head

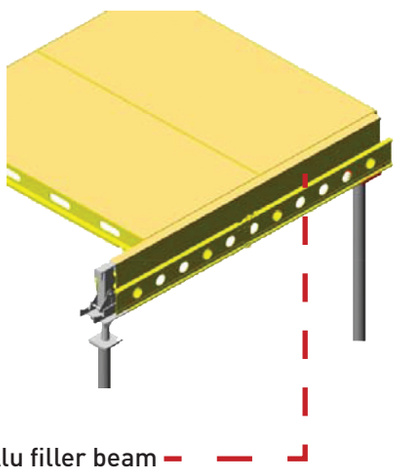


Prop head





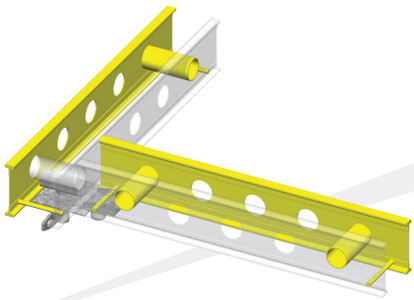
Fix the panels in both directions

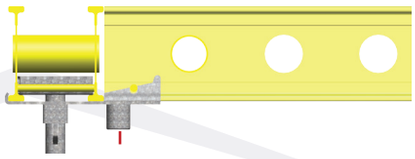


Alu filler beam is inserted between the drop heads.
Continous setting of panels without any fixing points between.

PMD Method Assembly

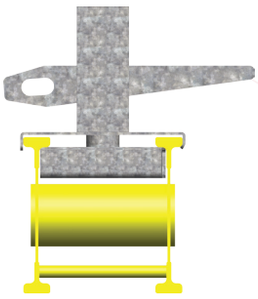
With the connection MB 90, it is possible to connect 2 main beams perpendicular on each other. Mainly used for open corner edges.

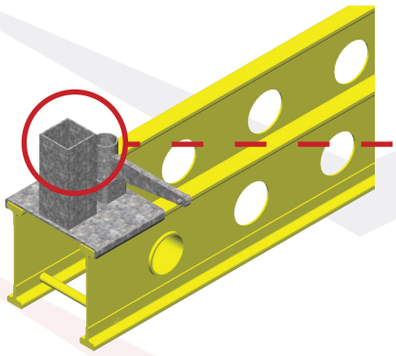




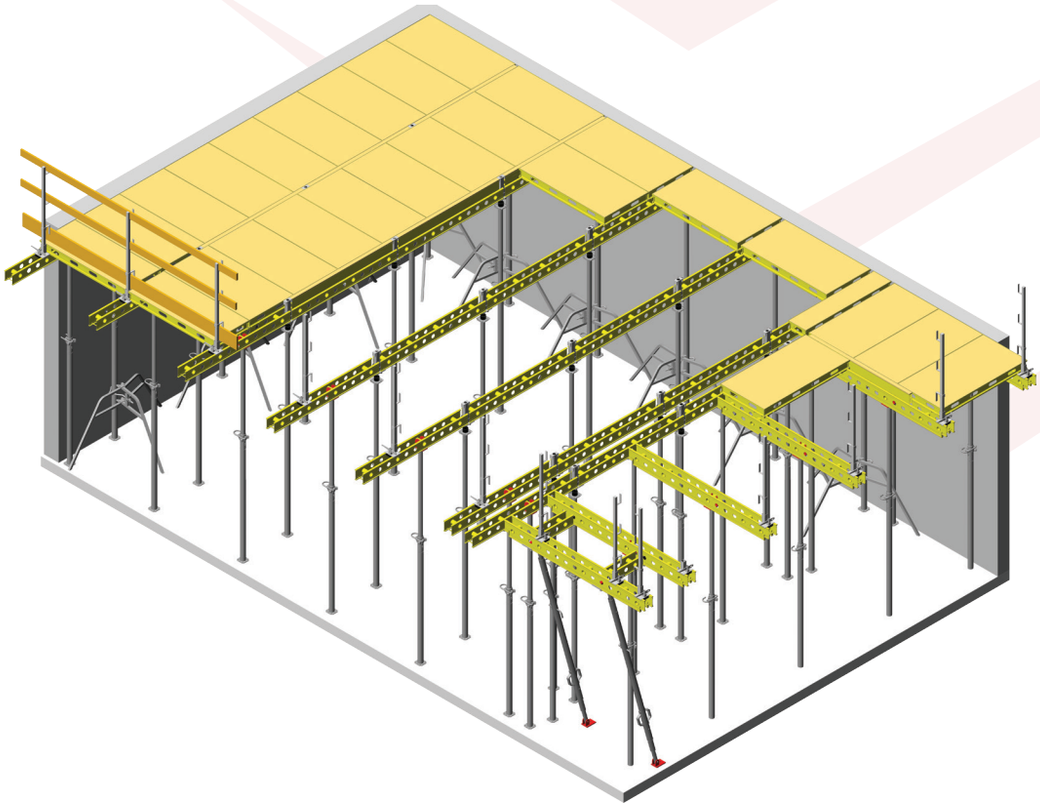
for fixing the prop to the main beam

For Safety, The Alu-deck end support for fixing handrails at the edges of the building.

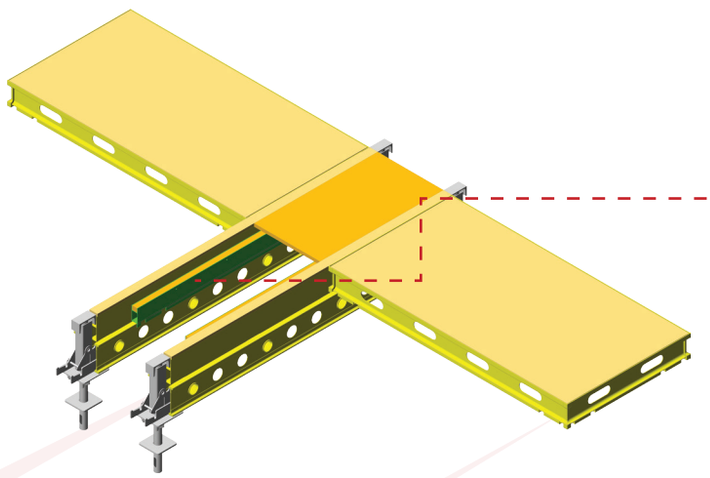




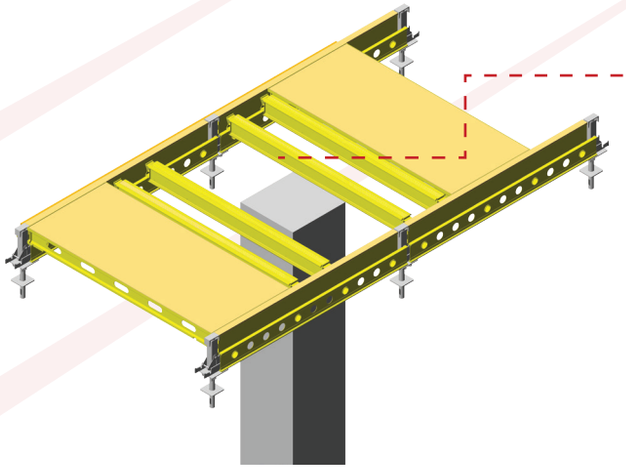
for fixing the handrail to the end support.



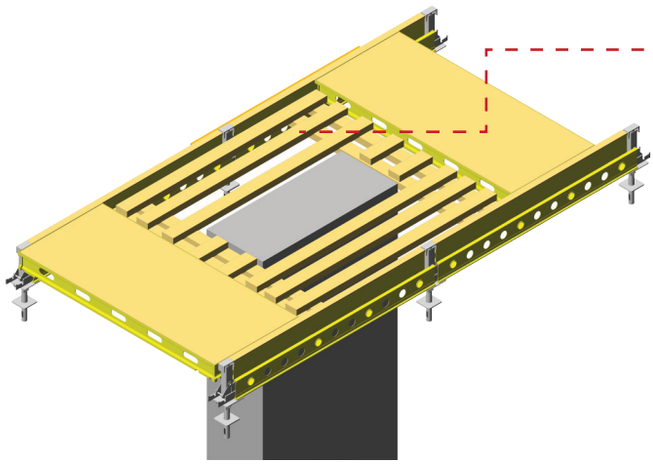
PMD Method Filler Solutions With 18 mm Plywood



Filler with compensation beams



Filler with infill beams



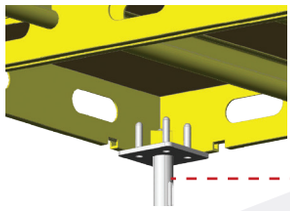
Filler with timber 2 x 4

Panel Method

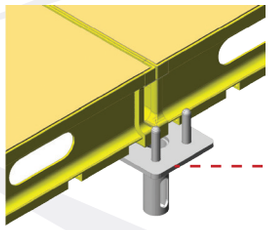
Only panels and panel-heads.

The self-feeding panels are directly supported by a panel-head.

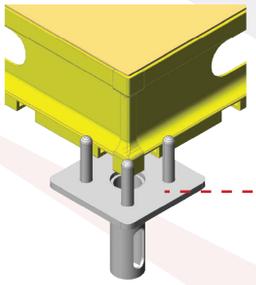
The crown head is used:
•In the cross point of 4 panels panels.
•To support 2 panels on a wall.
•Even in the corner to start the system.



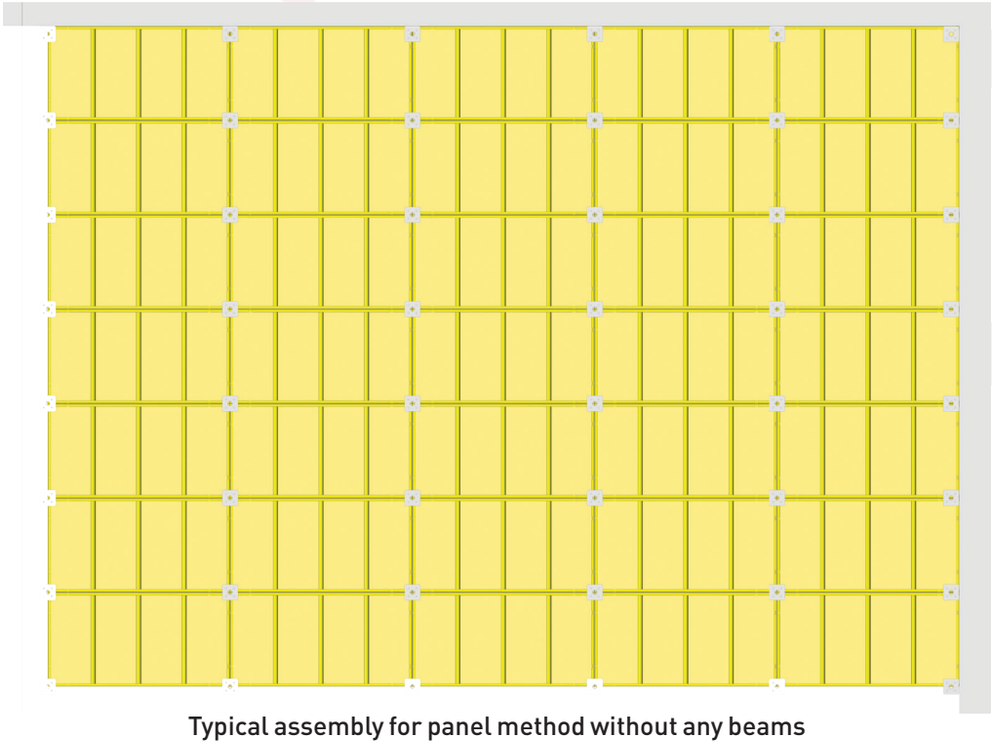
for corner support



for supporting 2 panels



cross point of 4 panels



Typical assembly for panel method without any beams

Design tables

These tables allow for a live load of 20% of the self-weight of the fresh concrete, but not less than 1.5 kN/m².

Distance between main beams 75 cm		Load on prop [kN]			
Slab thickness [cm]	Total load [kN/m^2]	Main-beam 120	Main-beam 150	Main-beam 180	Main-beam 240
12	4,9	4,4	5,5	6,6	8,8
14	5,4	5,0	6,1	7,3	9,7
16	5,9	5,4	6,6	8,0	10,6
18	6,4	5,9	7,2	8,6	11,5
20	7,0	6,5	7,9	9,5	12,6
22	7,5	6,9	8,4	10,1	13,5
24	8,0	9,4	9,0	10,8	14,4
26	8,5	7,9	9,6	11,5	15,3
28	9,0	8,3	10,1	12,2	16,2
30	9,6	8,9	10,8	13,0	17,3
32	10,0	9,2	11,2	13,5	18,0
34	10,6	9,8	11,9	14,3	19,1
36	11,2	10,3	12,6	15,2	20,2
38	11,9	11,0	13,3	16,0	21,3
40	12,7	11,7	14,3	17,1	22,9
50	15,9	14,7	17,9	21,5	28,6

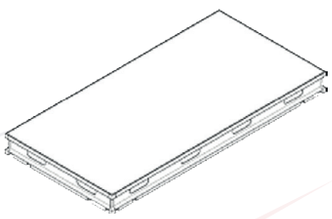
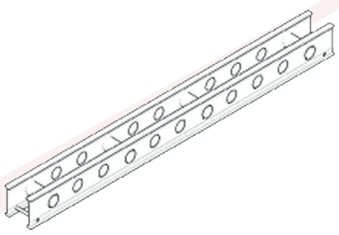
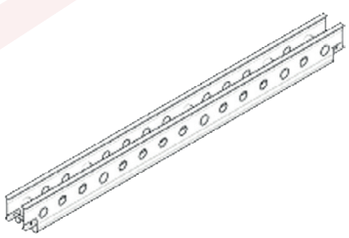
Distance between main beams 150 cm		Load on prop [kN]			
Slab thickness [cm]	Total load [kN/m^2]	Main-beam 120	Main-beam 150	Main-beam 180	Main-beam 240
12	4,9	8,8	11,0	13,2	17,6
14	5,4	9,7	12,2	14,6	19,4
16	5,9	10,6	13,3	15,9	21,2
18	6,4	11,5	14,4	17,3	23,0
20	7,0	12,6	15,8	18,9	25,2
22	7,5	13,5	16,9	20,3	27,0
24	8,0	14,4	18,0	21,6	28,8
26	8,5	14,3	19,1	23,0	30,6
28	9,0	16,2	20,3	24,3	32,4
30	9,6	17,3	21,6	25,9	34,6
32	10,0	18,0	22,5	27,0	35,9
34	10,6	19,1	23,9	28,6	38,2
36	11,2	20,2	25,3	30,3	

Design tables

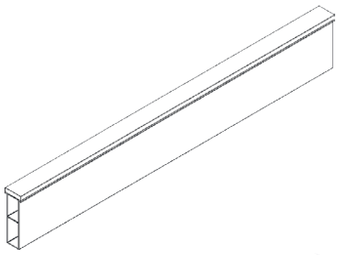
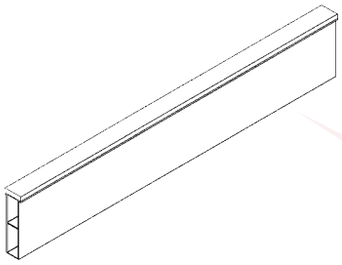
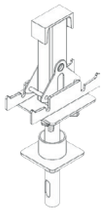
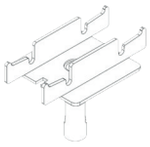
Distance between main beams 180 cm		Load on prop [kN]			
Slab thickness [cm]	Total load [kN/m^2]	Main-beam 120	Main-beam 150	Main-beam 180	Main-beam 240
12	4,9	10,6	13,2	15,9	21,2
14	5,4	11,7	14,6	17,5	23,3
16	5,9	12,7	15,9	19,1	25,5
18	6,4	13,9	17,3	20,7	27,6
20	7,0	15,1	18,9	22,7	30,2
22	7,5	16,2	20,3	24,3	32,4
24	8,0	17,3	21,6	25,9	34,6
26	8,5	18,4	23,0	27,5	36,7
28	9,0	19,5	24,3	29,2	38,9
30	9,6	20,8	25,9	31,1	
32	10,0	21,6	27,0	32,3	
34	10,6	22,9	28,6	34,3	
36	11,2	24,2	30,3	36,4	

Panel method		Load on prop [kN]	
Slab thickness [cm]	Total load [kN/m^2]	Panel 150 x 75	Panel 180 x 60
12	4,9	5,5	5,3
14	5,4	6,1	5,8
16	5,9	6,6	6,4
18	6,4	7,2	6,9
20	7,0	7,9	7,6
22	7,5	8,4	8,1
24	8,0	9,0	8,6
26	8,5	9,6	9,2
28	9,0	10,1	9,7
30	9,6	10,8	10,4
32	10,0	11,2	
34	10,6	11,9	
36	11,2	12,6	

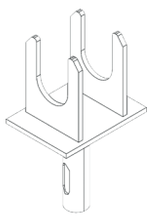
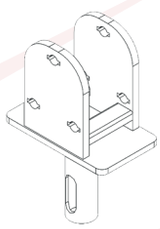
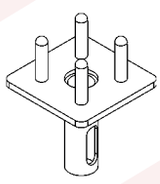
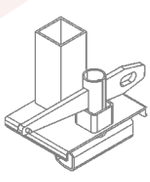
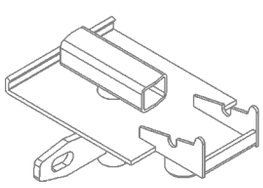
Components

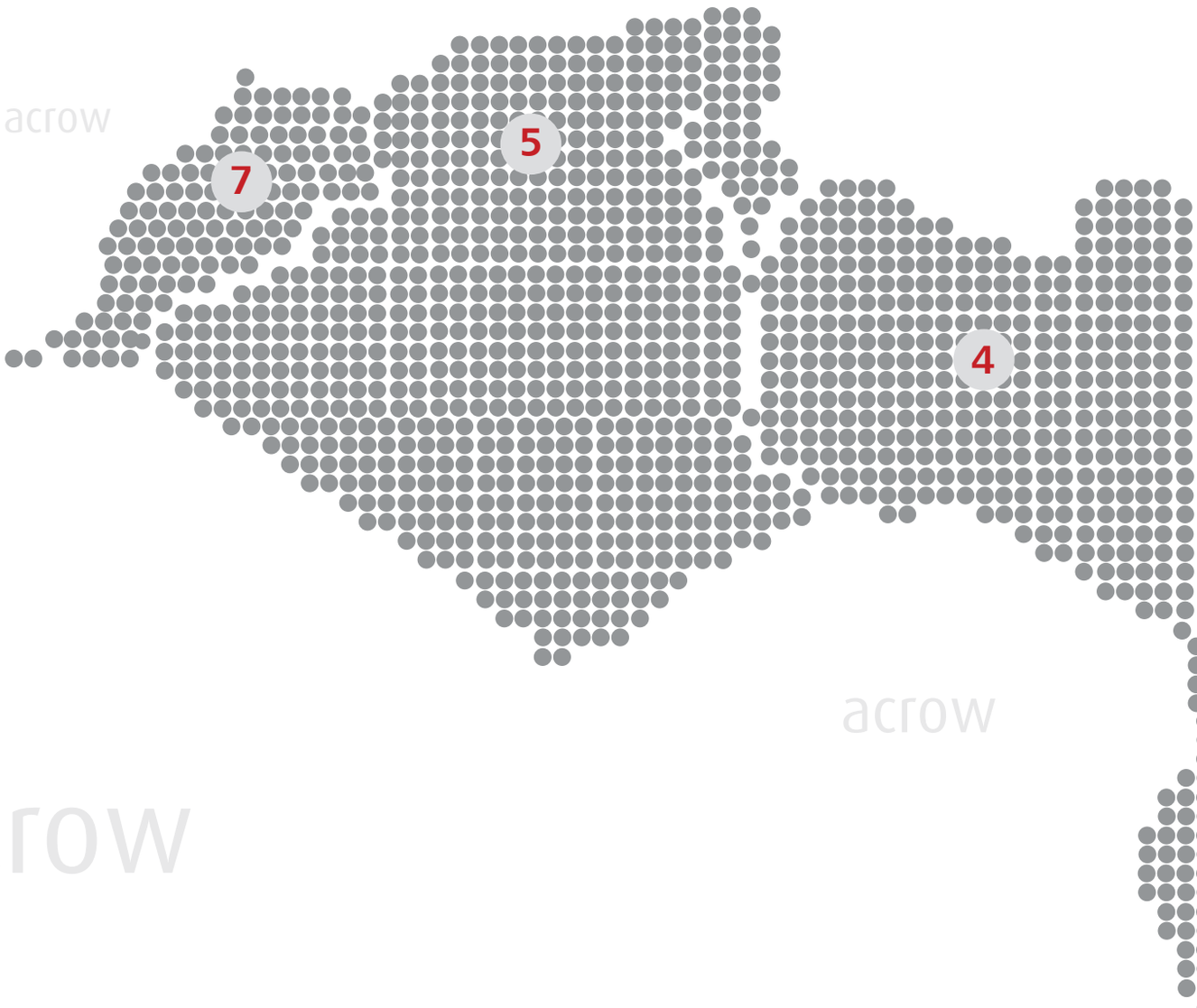
Item	Description	Code no.	Weight (kg)
	Panel ADB ADP 75 x 60 ADP 75 x 75 ADP 90 x 30 ADP 90 x 60 ADP 150 x 30 ADP 150 x 60 ADP 150 x 75 ADP 180 x 30 ADP 180 x 60	6221177062118 6221177062101 6221177062095 6221177062088 6221177062071 6221177062064 6221177062514 6221177062057 6221177062040	8.52 10.38 6.98 9.58 10.80 15.64 18.32 12.90 19.15
	Main Beam ADMB 60 ADMB 90 ADMB 120 ADMB 150 ADMB 180	6221177062200 6221177062194 6221177062187 6221177062170 6221177062163	3.37 4.88 6.58 8.3 9.8
	Main Beam ADMB 210 ADMB 225 ADMB 240 ADMB 360	6221177062156 6221177062149 6221177062132 6221177062125	15.2 16.23 17.47 26.18
	Infill Beams ADIB 90 ADIB 120 ADIB 150 ADIB 180	6221177062248 6221177062231 6221177062224 6221177062217	4.86 6.66 8.45 10.25

Components

Item	Description	Code no.	Weight (kg)
	ALU -Filler Beam 151/Lm AFB 151/0.50 m AFB 151/0.80 m AFB 151/1.10 m AFB 151/1.40 m AFB 151/1.70 m	 6221177062385 6221177062378 6221177062361 6221177062354 6221177062347	 1.35 2.16 2.97 3.79 4.60
	ALU -Filler Beam 176/Lm AFB 176/2.00m AFB 176/2.15 m AFB 176/2.30 m AFB 176/3.50 m	 6221177062323 6221177062316 6221177062309 6221177062293	 5.97 6.42 2.87 10.45
	Alu-Deck Drop head	6221177062255	8.92
	Alu-Deck Prop head	6221177080235	2.76

Components

Item	Description	Code no.	Weight (kg)
	Alu-Deck Beam head	6221177058685	2.57
	Alu-Deck Twin head	6221177080266	2.76
	Alu-Deck Panel head	6221177080259	1.45
	Alu-Deck End support	6221177058432	2.5
	Alu-Deck MB90 connection	6221177058425	2.6



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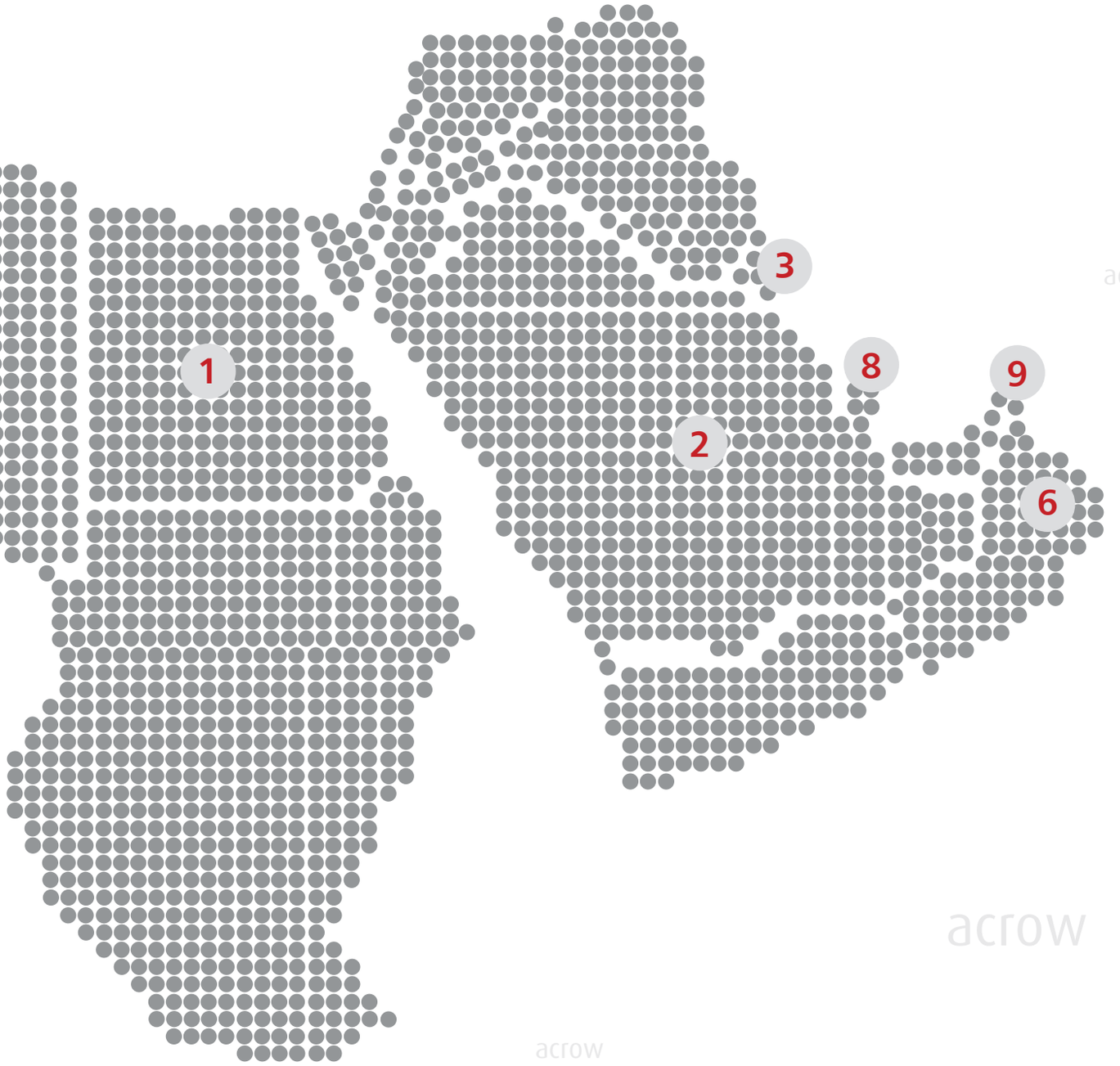
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Acrow Bridge table form


Acrow Circular column

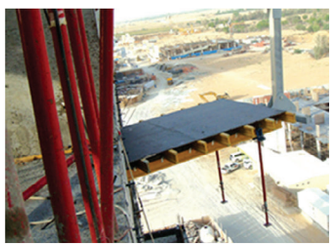

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